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Was Store K's profit last month at least 10 percent greater than its profit the previous month?

1. Store K's expenses last month were 5 percent greater than its expenses the previous month.
2. Store K's revenues last month were 10 percent greater than its revenues the previous month.

Algebra Applied Problems

Let P_{last}, E_{last}, and R_{last} be, respectively, the profit, expenses, and revenues for last month. Also, let P_{previous}, E_{previous}, and R_{previous} be, respectively, the profit, expenses, and revenues for the previous month. Then we have $P_{\text{last}} = R_{\text{last}} - E_{\text{last}}$ and $P_{\text{previous}} = R_{\text{previous}} - E_{\text{previous}}$. Determine whether $P_{\text{last}} \geq 1.1P_{\text{previous}}$ is true, or equivalently, determine whether $R_{\text{last}} - E_{\text{last}} \geq 1.1R_{\text{previous}} - 1.1E_{\text{previous}}$ is true.

1. Given that $E_{\text{last}} = 1.05E_{\text{previous}}$, it follows that the inequality $R_{\text{last}} - E_{\text{last}} \geq 1.1R_{\text{previous}} - 1.1E_{\text{previous}}$ is equivalent to $R_{\text{last}} - 1.1R_{\text{previous}} \geq -0.05E_{\text{previous}}$. It is clear that, for suitable values of R_{last}, R_{previous}, and E_{previous}, this last inequality could be true and this last inequality could be false; NOT sufficient.
2. Given that $R_{\text{last}} = 1.1R_{\text{previous}}$, it follows that the inequality $R_{\text{last}} - E_{\text{last}} \geq 1.1R_{\text{previous}} - 1.1E_{\text{previous}}$ is equivalent to $E_{\text{last}} \leq 1.1E_{\text{previous}}$. It is clear that, for suitable values of E_{last} and E_{previous}, this last inequality could be true and this last inequality could be false; NOT sufficient.

Taking (1) and (2) together, the following shows that $P_{\text{last}} \geq 1.1P_{\text{previous}}$ is true.

$$\begin{aligned} P_{\text{last}} &= R_{\text{last}} - E_{\text{last}} \\ &= 1.1R_{\text{previous}} - 1.05E_{\text{previous}} \\ &\geq 1.1R_{\text{previous}} - 1.1E_{\text{previous}} \end{aligned}$$

Therefore, $P_{\text{last}} \geq 1.1(R_{\text{previous}} - E_{\text{previous}})$, and hence $P_{\text{last}} \geq 1.1P_{\text{previous}}$ is true. (The reason for using ≥ above instead of > is to allow for the possibility that $E_{\text{previous}} = 0$.)

The correct answer is C; both statements together are sufficient.

•

Yesterday, Diana spent a total of 240 minutes attending a training class, responding to emails, and talking on the phone. If she did no two of these three activities at the same time, how much time did she spend talking on the phone?

1. Yesterday, the amount of time that Diana spent attending the training class was 90 percent of the amount of time that she spent responding to emails.
2. Yesterday, the amount of time that Diana spent attending the training class was 60 percent of the total amount of time that she spent responding to emails and talking on the phone.

Arithmetic Percents

Let T, E, and P be the number of minutes, respectively, that Diana spent attending the training class, responding to emails, and talking on the phone. We are given that $T + E + P = 240$ and are to determine the value of P.

1. Given that $T = 0.9E$, it is not possible to determine the value of P. Note that more than one appropriate value is possible for E (for example, $E = 50$ and $E = 100$), and each of these values leads to a possible value of T (for example, $T = 0.9 \times 50 = 45$ and $T = 0.9 \times 100 = 90$), after which different possible values of P can be found (for example, $P = 240 - 50 - 45 = 145$ and $P = 240 - 100 - 90 = 50$); NOT sufficient.
2. Given that $T = 0.6(E + P)$, it is not possible to determine the value of P. Note that using $T + E + P = 240$, we have $0.6(E + P) + (E + P) = 240$, or $1.6(E + P) = 240$, and hence $E + P = 150$. From this it follows that $T = 240 - (E + P) = 240 - 150 = 90$, but it is still not possible to determine the value of P since only $E + P = 150$ is required for the values of E and P. For example, it is possible that $T = 90$, $E = 50$, and $P = 100$, and it is possible that $T = 90$, $E = 100$, and $P = 50$; NOT sufficient.

Taking (1) and (2) together, we have $T = 0.9E$, $E + P = 150$, and $T = 90$. Therefore, $0.9E = 90$, or $E = 100$, and hence from $E + P = 150$, it follows that $100 + P = 150$, or $P = 50$.

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The correct answer is C; both statements together are sufficient.

•

When the wind speed is 9 miles per hour, the wind-chill factor w is given by

$$w = -17.366 + 1.19t,$$

where t is the temperature in degrees Fahrenheit. If at noon yesterday the wind speed was 9 miles per hour, was the wind-chill factor greater than 0?

1. The temperature at noon yesterday was greater than 10 degrees Fahrenheit.
2. The temperature at noon yesterday was less than 20 degrees Fahrenheit.

Algebra Applied Problems

Determine whether $-17.366 + 1.19t$ is greater than 0.

1. Given that $t > 10$, it follows that $-17.366 + 1.19t > -17.366 + 1.19(10)$, or $-17.366 + 1.19t > -5.466$. However, it is not possible to determine whether $-17.366 + 1.19t$ is greater than 0. For example, if $t = 19$, then $-17.366 + 1.19t = 5.244$ is greater than 0. However, if $t = 11$, then $-17.366 + 1.19t = -4.276$, which is not greater than 0; NOT sufficient.
2. Given that $t < 20$, the same examples used in (1) show that it is not possible to determine whether $-17.366 + 1.19t$ is greater than 0; NOT sufficient.

Taking (1) and (2) together is of no more help than either (1) or (2) taken separately because the same examples were used in both (1) and (2).

The correct answer is E; both statements together are still not sufficient.

•

How many members of a certain legislature voted against the measure to raise their salaries?

1. $\frac{1}{4}$ of the members of the legislature did not vote on the measure.
2. If 5 additional members of the legislature had voted against the measure, then the fraction of members of the legislature voting against the measure would have been $\frac{1}{3}$.

Arithmetic Ratio and Proportion

The task in this question is to determine whether, on the basis of statements 1 and 2, it is possible to calculate the number of members of the legislature who voted against a certain measure.

1. This statement, that $\frac{1}{4}$ of the members of the legislature did not vote on the measure, is compatible with any number of members of the legislature voting against the measure. After all, any number among the $\frac{3}{4}$ of the remaining members could have voted against the measure. Furthermore, based on statement 1, we do not know the number of members of the legislature (although we do know, based on this statement, that the number of members of the legislature is divisible by 4); NOT sufficient.
2. This statement describes a scenario, in which 5 additional members of the legislature vote against the measure and stipulates that $\frac{1}{3}$ of the members of the legislature would have voted against the measure in the scenario. Given this condition, we know that the number of members of the legislature was divisible by 3 and that the legislature had at least 15 members (to allow for the "5 additional members of the legislature" that could have voted against the measure, for a total of $\frac{1}{3}$ of the members voting against it). However, beyond this we know essentially nothing from statement 2. In particular, depending on the number of members of the legislature (which we have not been given), any number of members could have voted against the measure. For example, exactly one member could have voted against the measure, in which case the legislature would have had (1 + 5) × 3 = 18 members. Exactly two members could have voted against the measure, in which case the legislature

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× 3 = 18 members. Exactly two members could have voted against the measure, in which case the legislature would have had (2 + 5) × 3 = 21 members, and so on for 3 members voting against, 4 members voting against, and so on; NOT sufficient.

Considering the statements 1 and 2 together, the reasoning is similar to the reasoning for statement 2, but with the further condition that the total number of members of the legislature is divisible by 12 (so as to allow that both exactly $\frac{1}{4}$ of the members did not vote on the measure while exactly $\frac{1}{3}$ could have voted against the measure). For example, it could have been the case that the legislature had 24 members. In this case, $\frac{1}{3}$ of the members would have been 8 members, and, consistent with statements 1 and 2, 3 of the members (8 − 5) could have voted against the measure. Or the legislature could have had 36 members, in which case, consistent with statements 1 and 2, $\frac{1}{3}(36) - 5$ = 12 − 5 = 7 members could have voted against the measure.

The correct answer is E; both statements together are still not sufficient.

•

During a certain bicycle ride, was Sherry's average speed faster than 24 kilometers per hour? (1 kilometer = 1,000 meters)

1. Sherry's average speed during the bicycle ride was faster than 7 meters per second.
2. Sherry's average speed during the bicycle ride was slower than 8 meters per second.

Arithmetic Applied Problems

This problem can be solved by converting 24 kilometers per hour into meters per second. First, 24 kilometers is equivalent to 24,000 meters and 1 hour is equivalent to 3,600 seconds. Then, traveling 24 kilometers in 1 hour is equivalent to traveling 24,000 meters in 3,600 seconds, or $\frac{24{,}000}{3{,}600} = 6\frac{2}{3}$ meters per second.

1. This indicates that Sherry's average speed was faster than 7 meters per second, which is faster than $6\frac{2}{3}$ meters per second and, therefore, faster than 24 kilometers per hour; SUFFICIENT.
2. This indicates that Sherry's average speed was slower than 8 meters per second. Her average speed could have been 7 meters per second (since 7 < 8), in which case her average speed was faster than $6\frac{2}{3}$ meters per second and, therefore, faster than 24 kilometers per hour. Or her average speed could have been 5 meters per second (since 5 < 8), in which case her average speed was not faster than $6\frac{2}{3}$ meters per second and, therefore, not faster than 24 kilometers per hour; NOT sufficient.

The correct answer is A; statement 1 alone is sufficient.

•

At a two-day seminar, 90 percent of those registered attended the seminar on the first day. What percent of those registered did not attend the seminar on either day?

1. A total of 1,000 people registered for the two-day seminar.
2. Of those registered, 80 percent attended the seminar on the second day.

Arithmetic Statistics

1. Given that 1,000 people registered, the percent of those registered who did not attend the seminar on either day could be as low as 0% (if 1,000 attended the second day), and the percent could be as high as 10% (if none attended the second day); NOT sufficient.
2. Given that 80 percent of those registered attended the seminar on the second day, the Venn diagram below shows that the percent of those registered who did not attend the seminar on either day could be 0% (choose $N = 70$), and the percent could be 10% (choose $N = 80$); NOT sufficient.

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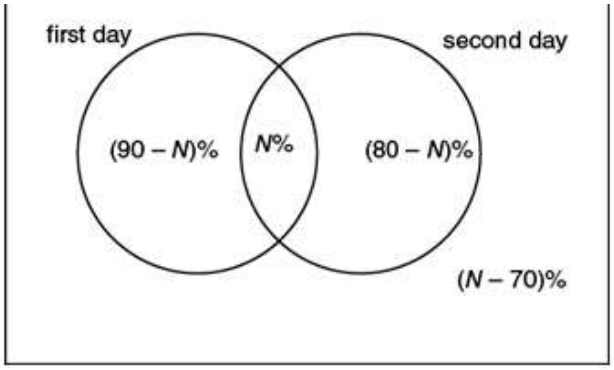

Taking (1) and (2) together, it is still not possible to determine the percent of those registered who did not attend the seminar on either day. This can be seen by modifying the Venn diagram above so that $(90 - N)\%$ is replaced with $900 - N$, $N\%$ is replaced with N, $(80 - N)\%$ is replaced with $800 - N$, and $(N - 70)\%$ is replaced with $N - 700$. Using the modified Venn diagram, it can be seen that the percent of those registered who did not attend the seminar on either day could be 0% (choose $N = 700$), and the percent could be 10% (choose $N = 800$).

The correct answer is E; both statements together are still not sufficient.

•

Yesterday between 9:00 a.m. and 6:00 p.m. at Airport X, all flights to Atlanta departed at equally spaced times and all flights to New York City departed at equally spaced times. A flight to Atlanta and a flight to New York City both departed from Airport X at 1:00 p.m. yesterday. Between 1:00 p.m. and 3:00 p.m. yesterday, did another pair of flights to these 2 cities depart from Airport X at the same time?

1. Yesterday at Airport X, a flight to Atlanta and a flight to New York City both departed at 10:00 a.m.
2. Yesterday at Airport X, flights to New York City departed every 15 minutes between 9:00 a.m. and 6:00 p.m.

Arithmetic Applied Problems

It is useful to note that although the departures discussed all lie between 9:00 a.m. and 6:00 p.m., there is no information concerning when the first departures took place during this time other than what is necessary for the information to be consistent. For example, since departures to both Atlanta and New York City took place at 1:00 p.m., the first departure to either of these cities could not have occurred after 1:00 p.m.

1. Given that departures to both Atlanta and New York City took place at 10:00 a.m., it is not possible to determine whether simultaneous departures to these cities occurred between 1:00 p.m. and 3:00 p.m. For example, it is possible that departures to both Atlanta and New York City took place every 15 minutes beginning at 9:15 a.m., and thus it is possible that simultaneous departures to both these cities occurred between 1:00 p.m. and 3:00 p.m. However, it is also possible that departures to Atlanta took place every 3 hours beginning at 10:00 a.m. and departures to New York City took place every 15 minutes beginning at 9:15 a.m., and thus it is possible that no simultaneous departures to these cities occurred between 1:00 p.m. and 3:00 p.m.; NOT sufficient.
2. Given that departures to New York City took place every 15 minutes, the same examples used in (1) can be used to show that it is not possible to determine whether simultaneous departures to these cities occurred between 1:00 p.m. and 3:00 p.m.; NOT sufficient.

Taking (1) and (2) together, it is still not possible to determine whether simultaneous departures to these cities occurred between 1:00 p.m. and 3:00 p.m. because both (1) and (2) are true for the examples above.

The correct answer is E; both statements together are still not sufficient.

•

A bank account earned 2% annual interest, compounded daily, for as long as the balance was under $1,000, starting when the account was opened. Once the balance reached $1,000, the account earned 2.5% annual interest, compounded daily, until the account was closed. No deposits or withdrawals were made. Was the total amount of interest earned at the 2% rate greater than the total amount earned at the 2.5% rate?

1. The account earned exactly $25 in interest at the 2.5% rate.
2. The account was open for exactly 3 years.

Arithmetic Percents

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1. We are given that the account earned exactly $25 in interest at the 2.5% rate. If the account was opened with $999.99, then after one day, the balance would have reached $1,000 because the amount of interest earned in this day would be almost exactly $(\$1{,}000)\left(\frac{0.025}{365}\right) = \$\left(\frac{25}{365}\right) = \$\left(\frac{5}{73}\right)$, which is greater than $\$\left(\frac{5}{100}\right) = \$0.05$. However, the amount of interest earned in this day would be less than $25. Therefore, the amount of interest earned at the 2% rate would be less than the amount of interest earned at the 2.5% rate. On the other hand, if the account was opened with $950, then the account would have earned at least $50 in interest at the 2% rate, which is greater than $25. Therefore, the amount of interest earned at the 2% rate would be greater than the amount of interest earned at the 2.5% rate; NOT sufficient.
2. We are given that the account was open for exactly 3 years. Depending on the amount the account was opened with, the interest at the 2% rate could have been earned in only a few days, or in all but a few days less than 3 years. In the former case, the amount of interest earned at the 2% rate would be less than the amount of interest earned at the 2.5% rate, and in the latter case, the amount of interest earned at the 2% rate would be greater than the amount of interest earned at the 2.5% rate; NOT sufficient.

Taking (1) and (2) together, the account was open for 3 years and earned $25 in interest at the 2.5% rate. Since $25 in **simple** interest would be earned on $1,000 at a 2.5% rate for 1 year, it follows that more than $25 in **compound** interest would be earned on $1,000 (or a greater amount) at a 2.5% rate compounded daily for 1 year. Therefore, the $25 in interest earned at the 2.5% rate was earned in less than 1 year, and hence the interest at the 2% rate was earned in more than 2 years. If the interest earned at the 2% rate was less than or equal to $25, then the account would have been opened with at least $975. However, at a 2% simple interest rate, $975 would earn ($975)(0.02) = $19.50 in 1 year, and hence more than (2)($19.50) = $39 in more than 2 years, which contradicts the assumption that the interest earned at the 2% rate was less than or equal to $25. Therefore, the interest earned at the 2% rate must have been greater than $25, and hence the interest earned at the 2% rate was greater than the interest earned at the 2.5% rate.

The correct answer is C; both statements together are sufficient.

•

What is the average (arithmetic mean) annual salary of the 6 employees of a toy company?

1. If the 6 annual salaries were ordered from least to greatest, each annual salary would be $6,300 greater than the preceding annual salary.
2. The range of the 6 annual salaries is $31,500.

Arithmetic Statistics

Can we determine the arithmetic mean of the annual salaries of the 6 employees?

1. Given only that the 6 annual salaries can be put into a sequence from least to greatest, with a difference of $6,300 between adjacent members of the sequence, we can infer certain things about the mean of the salaries. For example, because none of the salaries would be negative, we know from statement 1 that the mean of the salaries is greater than or equal to $\frac{0 + \$6{,}300 + \$12{,}600 + \$18{,}900 + \$25{,}200 + \$31{,}500}{6}$. (It is not necessary to perform this calculation.) However, depending on what the least of the salaries is—that is, the value at which the sequence of salaries begins—the average of the salaries could, consistent with condition 1, take on any value greater than this quotient; NOT sufficient.
2. Given the statement that the range of the salaries is $31,500, reasoning similar to the reasoning for statement 1 applies. A difference between least salary and greatest salary of $31,500 is consistent with any value for the least salary, so long as the greatest salary is $31,500 greater than the least salary. Furthermore, even if we knew what the least and the greatest salaries are, it would be impossible to determine the mean merely from the range; NOT sufficient.

As reflected in the numerator of the quotient in the discussion of statement 1, we can see that statement 1 implies statement 2. In the sequence of 6 salaries with a difference of $6,300 between adjacent members of the sequence, the difference between the least salary and the greatest salary is 5 × $6,300 = $31,500. Therefore, because statement 1 is insufficient for determining the mean of the salaries, the combination of statement 1 and statement 2 is also insufficient for determining the mean of the salaries.

The correct answer is E; both statements together are not sufficient.

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The correct answer is E; both statements together are not sufficient.

•

Each person attending a fundraising party for a certain club was charged the same admission fee. How many people attended the party?

1. If the admission fee had been \$0.75 less and 100 more people had attended, the club would have received the same amount in admission fees.
2. If the admission fee had been \$1.50 more and 100 fewer people had attended, the club would have received the same amount in admission fees.

Algebra Simultaneous Equations

Let $\$p$ be the admission fee that each of the n people paid. Then the total of the admission fees is $\$pn$. We are to determine the value of n.

1. We are given that if the admission fee had been $\$(p-0.75)$ and $(n+100)$ people had attended, then the total of the admission fees would have been $\$pn$, or equivalently, $(p-0.75)(n+100)=pn$. Expanding the left side of this equation gives $pn-0.75n+100p-75=pn$, or equivalently, $100p=0.75n+75$. However, it is not possible to determine the value of n because we can choose different positive integers for n and solve for the resulting values of p. For example, $n=100$ and $p=1.5$ is possible, and $n=900$ and $p=7.5$ is possible; NOT sufficient.
2. We are given that if the admission fee had been $\$(p+1.5)$ and $(n-100)$ people had attended, then the total of the admission fees would have been $\$pn$, or equivalently, $(p+1.5)(n-100)=pn$. Expanding the left side of this equation gives $pn+1.5n-100p-150=pn$, or equivalently, $100p=1.5n-150$. However, it is not possible to determine the value of n because we can choose different positive integers for n and solve for the resulting values of p. For example, $n=200$ and $p=1.5$ is possible, and $n=1{,}100$ and $p=15$ is possible; NOT sufficient.

Taking (1) and (2) together, it follows that $100p=0.75n+75$ and $100p=1.5n-150$. Therefore, $0.75n+75=1.5n-150$, which can be solved for a unique value of n, namely $n=300$.

The correct answer is C; both statements together are sufficient.

•

A total of 20 amounts are entered on a spreadsheet that has 5 rows and 4 columns; each of the 20 positions in the spreadsheet contains one amount. The average (arithmetic mean) of the amounts in row i is R_i $(1 \le i \le 5)$. The average of the amounts in column j is C_j $(1 \le j \le 4)$. What is the average of all 20 amounts on the spreadsheet?

1. $R_1+R_2+R_3+R_4+R_5=550$
2. $C_1+C_2+C_3+C_4=440$

Arithmetic Statistics

It is given that R_i represents the average of the amounts in row i. Since there are four amounts in each row, $4R_i$ represents the total of the amounts in row i. Likewise, it is given that C_j represents the average of the amounts in column j. Since there are five amounts in each column, $5C_j$ represents the total of the amounts in column j.

1. It is given that $R_1+R_2+R_3+R_4+R_5=550$, and so $4(R1+R_2+R_3+R_4+R_5)=4R_1+4R_2+4R_3+4R_4+4R_5=4(550)=2{,}200$. Therefore, 2,200 is the sum of all 20 amounts (4 amounts in each of 5 rows), and the average of all 20 amounts is $\frac{2{,}200}{20}=110$; SUFFICIENT.
2. It is given that $C_1+C_2+C_3+C_4=440$, and so $5(C_1+C_2+C_3+C_4)=5C_1+5C_2+5C_3+5C_4=5(440)=2{,}200$. Therefore, 2,200 is the sum of all 20 amounts (5 amounts in each of 4 columns), and the average of all 20 amounts is $\frac{2{,}200}{20}=110$; SUFFICIENT.

The correct answer is D; each statement alone is sufficient.

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•

Was the range of the amounts of money that Company Y budgeted for its projects last year equal to the range of the amounts of money that it budgeted for its projects this year?

1. Both last year and this year, Company Y budgeted money for 12 projects and the least amount of money that it budgeted for a project was $400.
2. Both last year and this year, the average (arithmetic mean) amount of money that Company Y budgeted per project was $2,000.

Arithmetic Statistics

Let G_1 and L_1 represent the greatest and least amounts, respectively, of money that Company Y budgeted for its projects last year, and let G_2 and L_2 represent the greatest and least amounts, respectively, of money that Company Y budgeted for its projects this year. Determine if the range of the amounts of money Company Y budgeted for its projects last year is equal to the range of amounts budgeted for its projects this year; that is, determine if $G_1 - L_1 = G_2 - L_2$.

1. This indicates that $L_1 = L_2 = \$400$, but does not give any information about G_1 or G_2; NOT sufficient.
2. This indicates that the average amount Company Y budgeted for its projects both last year and this year was $2,000 per project but does not give any information about the least and greatest amounts that it budgeted for its projects either year; NOT sufficient.

Taking (1) and (2) together, it is known that $L_1 = L_2 = \$400$ and that the average amount Company Y budgeted for its projects both last year and this year was $2,000 per project, but there is no information about G_1 or G_2. For example, if, for each year, Company Y budgeted $400 for each of 2 projects and $2,320 for each of the 10 others, then (1) and (2) are true and the range for each year was $2,320 − $400 = $1,920. However, if, last year, Company Y budgeted $400 for each of 2 projects and $2,320 for each of the 10 others, and, this year, budgeted $400 for each of 11 projects and $19,600 for 1 project, then (1) and (2) are true, but the range for last year was $1,920 and the range for this year was $19,600 − $400 = $19,200.

The correct answer is E; both statements together are still not sufficient.

•

What is the probability that Lee will make exactly 5 errors on a certain typing test?

1. The probability that Lee will make 5 or more errors on the test is 0.27.
2. The probability that Lee will make 5 or fewer errors on the test is 0.85.

Arithmetic Probability

1. Given that 0.27 is the probability that Lee will make 5 or more errors on the test, it is clearly not possible to determine the probability that Lee will make exactly 5 errors on the test; NOT sufficient.
2. Given that 0.85 is the probability that Lee will make 5 or fewer errors on the test, it is clearly not possible to determine the probability that Lee will make exactly 5 errors on the test; NOT sufficient.

Taking (1) and (2) together, let E be the event that Lee will make 5 or more errors on the test and let F be the event that Lee will make 5 or fewer errors on the test. Then $P(E \text{ or } F) = 1$, since it will always be the case that, when taking the test, Lee will make at least 5 errors or at most 5 errors. Also, (1) and (2) can be expressed as $P(E) = 0.27$ and $P(F) = 0.85$, and the question asks for the value of $P(E \text{ and } F)$. Using the identity $P(E \text{ or } F) = P(E) + P(F) - P(E \text{ and } F)$, it follows that $1 = 0.27 + 0.85 - P(E \text{ and } F)$, or $P(E \text{ and } F) = 0.27 + 0.85 - 1 = 0.12$. Therefore, the probability that Lee will make exactly 5 errors on the test is 0.12.

The correct answer is C; both statements together are sufficient.

•

Last month, Grace worked, and was paid for, a total of x hours. Some of the hours were on the day shift, and the remainder of the hours were on the night shift. Her hourly pay is 20% higher for the night shift than for the day shift. How many hours did Grace work on the day shift last month?

1. $x = 55$

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2. Grace's gross pay for the hours she worked on the day shift last month was exactly 50% of her total gross pay for last month.

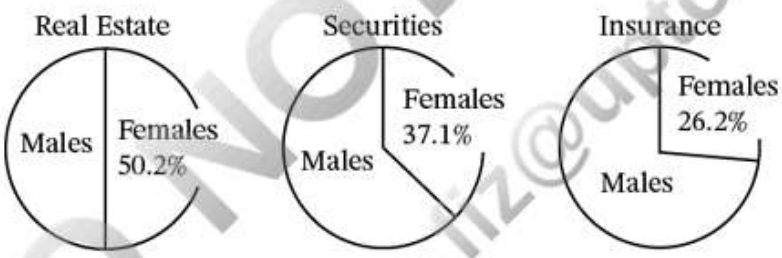
Algebra Percents

1. We are given that Grace worked a total of 55 hours. Thus, if Grace worked d hours on the day shift, then she worked $(55 - d)$ hours on the night shift. Although we know that Grace's hourly pay is 20% higher on the night shift, we do not know her hourly pay on either shift, nor is there any connection between her hourly pay on either shift and the number of hours she worked on either or both shifts. For example, Grace could have worked 30 hours on the day shift and 25 hours on the night shift, and Grace could have worked 25 hours on the day shift and 30 hours on the night shift, with each possibility having any hourly pay consistent with being 20% higher on the night shift; NOT sufficient.
2. We are given that Grace's total pay for the day shift was 50% of her total pay for both shifts combined. Therefore, Grace's total pay for the day shift was equal to her total pay for the night shift. However, any distribution of hours such that Grace worked 20% longer on the day shift than on the night shift is possible. For example, if Grace worked 24 hours at $\$p$ per hour on the day shift and 20 hours at $\$1.2p$ per hour on the night shift, then her total pay for the day shift would have been $\$24p$, and her total pay for the night shift would have been $(20)(\$1.2p) = \$24p$. On the other hand, if Grace worked 36 hours at $\$p$ per hour on the day shift and 30 hours at $\$1.2p$ per hour on the night shift, then her total pay for the day shift would have been $\$36p$, and her total pay for the night shift would have been $(30)(\$1.2p) = \$36p$; NOT sufficient.

Taking (1) and (2) together, letting d be the number of hours Grace worked on the day shift, and letting $\$p$ be her hourly rate on the day shift, it follows that $\$dp$ (her total pay for the day shift) is equal to $\$(55 - d)(1.2p)$ (her total pay for the night shift). Therefore, $dp = (55 - d)(1.2p)$. Dividing both sides by p gives $d = (55 - d)(1.2)$, which can be solved for a unique value of d, namely $d = 30$.

The correct answer is C; both statements together are sufficient.

DISTRIBUTION OF SALESPERSONS
BY GENDER IN THREE SECTORS, YEAR X



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In year X were there more female salespersons in the securities sector than in the insurance sector?

1. There were more male salespersons in the insurance sector than in the securities sector.
2. The total number of salespersons was greater in the securities sector than in the insurance sector.

Arithmetic Percents

From the graphs, 37.1% of the salespersons in securities are females and (100 − 37.1)% = 62.9% are males; 26.2% of the salespersons in insurance are females and (100 − 26.2)% = 73.8% are males. Let S and I represent the number of salespersons in securities and insurance, respectively. Let F_S, M_S, F_I, and M_I, represent the numbers of female and male salespersons, respectively, in securities and insurance, respectively. Determine if the inequality $F_S > F_I$ is true.

1. Given that $M_I > M_S$, it is not possible to determine if $F_S > F_I$ is true. For example, if $S = 10{,}000$ and $I = 9{,}000$, then $M_S = 0.629(10{,}000) = 6{,}290$, $M_I = 0.738(9{,}000) = 6{,}642$, $F_S = 0.371(10{,}000) = 3{,}710$, and $F_I = 0.262(9{,}000) = 2{,}358$. These numbers are summarized in the table below, from which it is easy to see that $M_I > M_S$ and $F_S > F_I$.

Sector	Number of employees	Males	Females

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	employees		
Securities	10,000	6,290	3,710
Insurance	9,000	6,642	2,358

However, if $S = 10{,}000$ and $I = 20{,}000$, then $M_S = 0.629(10{,}000) = 6{,}290$, $M_I = 0.738(20{,}000) = 14{,}760$, $F_S = 0.371(10{,}000) = 3{,}710$ and $F_I = 0.262(20{,}000) = 5{,}240$. These numbers are summarized in the table below, from which it is easy to see that $M_I > M_S$ and $F_S < F_I$; NOT sufficient.

Sector	Number of employees	Males	Females
Securities	10,000	6,290	3,710
Insurance	20,000	14,760	5,240

2. Given that $S > I$, it follows that $0.371S > 0.371I$ and $0.371I > 0.262I$, so $0.371S > 0.262I$ and $F_S > F_I$; SUFFICIENT.

The correct answer is B; statement 2 alone is sufficient.

•

If a club made a gross profit of $0.25 for each candy bar it sold, how many candy bars did the club sell?

1. The total revenue from the sale of the candy bars was $300.
2. If the club had sold 80 more candy bars, its gross profits would have increased by 20 percent.

Algebra First-Degree Equations

Let n be the number of candy bars sold. The gross profit from selling the n candy bars was $0.25n$. What is the value of n, or equivalently, what is the value of $0.25n$?

1. Given that the total revenue was $300, it is not possible to determine how many candy bars the club sold because nothing is known about the total cost, which is the value of $300 - 0.25n$; NOT sufficient.
2. Given that the gross profit from selling $(n + 80)$ candy bars is equal to $(1.2)(0.25n)$, it follows that $(0.25)(n + 80) = (1.2)(0.25n)$. Therefore, $0.25n + 20 = (1.2)(0.25n)$, or $20 = (0.2)(0.25n)$, and hence $0.25n = 100$; SUFFICIENT.

The correct answer is B; statement 2 alone is sufficient.

•

In one year 2,100 malpractice claims were filed with insurance company X and of these $\frac{1}{4}$ resulted in a financial settlement. Of those resulting in a financial settlement of less than $400,000, what was the average payment per claim?

1. Company X paid a total of 24.5 million dollars to the claimants.
2. Only 5 claims resulted in payments of $400,000 or more.

Arithmetic Statistics

A total of $\frac{1}{4}(2{,}100) = 525$ claims were paid. What was the average payment per claim of those claims having a payment less than $400,000?

1. Given that the total payment for the 525 paid claims was $24.5 million, it is not possible to determine the average payment per claim. This is because almost nothing is known about the total payment for those claims less than $400,000 or the number of those claims less than $400,000 (we only know they cannot exceed $24.5

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million and 525 paid claims), and thus, more than one value is possible for the average payment per claim for those claims less than $400,000; NOT sufficient.

2. Given that a total of 5 paid claims had payments of $400,000 or greater, it is not possible to determine the average payment per claim. This is because nothing is known about the total payment for those claims less than $400,000, and thus, more than one value is possible for this total payment divided by 520 (i.e., the average payment per claim for those claims less than $400,000); NOT sufficient.

Taking (1) and (2) together, it is still not possible to determine the average payment per claim. For example, if each of the 5 paid claims of over $400,000 was equal to $500,000, then the average payment per claim for those claims less than $400,000 would be $\frac{\$24.5 \text{ million} - \$2.5 \text{ million}}{520} = \frac{\$22 \text{ million}}{520}$. However, if each of the 5 paid claims of over $400,000 was equal to $4 million, then the average payment per claim for those claims less than $400,000 would be $\frac{\$24.5 \text{ million} - \$20 \text{ million}}{520} = \frac{\$4.5 \text{ million}}{520}$, which is different from the first example.

The correct answer is E; both statements together are still not sufficient.

•

If there are 13 boys in club X, what is the average age of these boys?

1. The oldest boy is 13 years old and the youngest boy is 9 years old.
2. Eleven of the boys are either 10 years old or 11 years old.

Arithmetic Statistics

What is the average age of the 13 boys, or equivalently, what is the sum of the ages of the 13 boys?

1. Given that the oldest boy is 13 and the youngest boy is 9, it is not possible to determine the sum of their ages. For example, if their ages were such that one is 9, eleven are 10, and one is 13, then the sum of their ages would be less than if their ages were such that one is 9, eleven are 11, and one is 13; NOT sufficient.
2. Given that eleven of the boys are either 10 or 11, it is not possible to determine the sum of their ages, because the same examples used above are such that eleven of the boys are either 10 or 11; NOT sufficient.

Taking (1) and (2) together, it is not possible to determine the sum of their ages because the examples above satisfy both (1) and (2).

The correct answer is E; both statements together are still not sufficient.

•

If all the employees of a company fall into one and only one of 3 groups, X, Y, or Z, with 250, 100, and 20 members in each group, respectively, what is the average (arithmetic mean) weekly salary of all the employees of this company, if all employees are paid every week of the year?

1. The average (arithmetic mean) annual salary of the employees in Group X is $10,000, in Group Y $15,000 and in Group Z $20,000.
2. The total annual payroll is $4,400,000.

Arithmetic Statistics

The average of the weekly salaries is the average of the annual salaries divided by 52, and thus the average of the weekly salaries can be determined if and only if the average of the annual salaries can be determined. What is the average of the annual salaries, or equivalently, what is the sum of the annual salaries?

1. Given that the average of the annual salaries of employees in Groups X, Y, and Z is $10,000, $15,000, and $20,000, respectively, it follows that the sum of the annual salaries of employees in Groups X, Y, and Z is 250($10,000), 100($15,000), and 20($20,000), respectively. Therefore, the sum of the annual salaries is the sum of these three amounts; SUFFICIENT.
2. We are given that the sum of the annual salaries is $4,400,000; SUFFICIENT.

The correct answer is D; each statement alone is sufficient.

DISTRIBUTION OF SALES INCOME
FOR STORE *S* LAST WEEK

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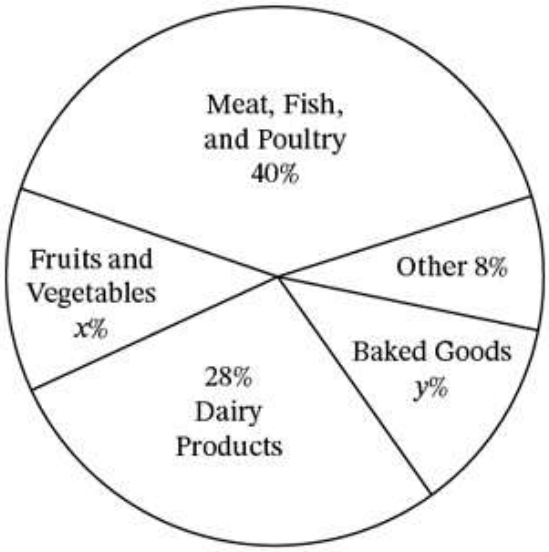

100% = $100,000

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According to the graph above, the sale of fruits and vegetables in Store S last week accounted for what percent of the total sales income for the week?

1. Last week the total income from the sale of fruits and vegetables in Store S was $16,000.
2. $x = 2y$

Algebra Percents

According to the graph, sales of fruits and vegetables accounted for $x\%$ of the total sales, baked goods accounted for $y\%$, and all other categories combined accounted for 76%. It follows that $x + y = 100 - 76 = 24$. Determine the value of x.

1. Given that sales of fruits and vegetables accounted for $x\%$ of the total sales of $100,000 and this amounted to $16,000, it follows that $\frac{x}{100}(100{,}000) = 16{,}000$. Thus, $x = 16$; SUFFICIENT.
2.

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